

## CURATED RESEARCH PAPERS

### Top 10 Research Papers on Agentic Frameworks for Runtime Self-Correction

#### 1. Self-Refine: Iterative Refinement with Self-Feedback

**Authors:** Madaan et al. (2023)

**Key Contribution:** Framework for iterative self-improvement using feedback

**Venue:** NeurIPS 2023

**Impact:** Introduces systematic approach to self-correction

#### 2. Reflexion: Language Agents with Verbal Reinforcement Learning

**Authors:** Shinn et al. (2023)

**Key Contribution:** Verbal reinforcement learning for agent improvement

**Venue:** ICLR 2023

**Impact:** Demonstrates learning from linguistic feedback

#### 3. Constitutional AI: Harmlessness from AI Feedback

**Authors:** Bai et al. (2022)

**Key Contribution:** Self-supervised approach to safety alignment

**Venue:** Anthropic Technical Report

**Impact:** Foundational work on AI self-correction for safety

#### 4. Tree of Thoughts: Deliberate Problem Solving with Large Language Models

**Authors:** Yao et al. (2023)

**Key Contribution:** Structured reasoning and self-evaluation

**Venue:** ICML 2023

**Impact:** Enhanced planning and error detection capabilities

#### 5. Self-Consistency Improves Chain of Thought Reasoning in Language Models

**Authors:** Wang et al. (2022)

**Key Contribution:** Multiple reasoning paths for error detection

**Venue:** ICLR 2022

**Impact:** Improved reliability through consistency checking

#### 6. Learning to Self-Correct via Reinforcement Learning

**Authors:** Welleck et al. (2023)

**Key Contribution:** RL-based framework for automatic correction

**Venue:** ICML 2023

**Impact:** Practical approach to learning self-correction

**7. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing**

**Authors:** Gou et al. (2023)

**Key Contribution:** Tool-augmented self-correction framework

**Venue:** ICLR 2024

**Impact:** Integration of external tools for verification

**8. Self-Debug: Teaching Large Language Models to Debug Their Predicted Program**

**Authors:** Chen et al. (2023)

**Key Contribution:** Code debugging through self-reflection

**Venue:** ICLR 2024

**Impact:** Domain-specific self-correction for programming

**9. Teaching Language Models to Self-Debug**

**Authors:** Gou et al. (2023)

**Key Contribution:** Systematic approach to error identification

**Venue:** EMNLP 2023

**Impact:** Generalizable debugging methodology

**10. Self-Taught Optimizer (STOP): Recursively Self-Improving Code Generation**

**Authors:** Zelikman et al. (2023)

**Key Contribution:** Recursive self-improvement for code generation

**Venue:** ICLR 2024

**Impact:** Demonstrates continuous self-enhancement